TEXORA · COURSE NOTES · LECTURE SHEET

Matrices

Arrays of numbers and the algebra that works on them — types, operations, determinants, inverses, rank, and the solution of linear systems.

MATH-1101 · MATHEMATICS-I · TOPIC 1 · LEVEL 1 TERM 1

1. What a matrix is

A **matrix** is a rectangular array of numbers (or symbols) arranged in rows and columns, enclosed in brackets. A matrix with *m* rows and *n* columns is of **order m × n**, and its entries are written a_{ij}, where *i* names the row and *j* the column. Matrices are the working language of every subject that handles many quantities at once — systems of equations in engineering, stress and strain in mechanics, and data tables everywhere. Two matrices are **equal** only when they have the same order and every corresponding entry is equal.

1.1 The standard types

TYPE	DEFINITION
Row / column matrix	A single row (1 × n) or a single column (m × 1).
Square matrix	m = n; the entries a_{11}, a_{22}, … form the *principal diagonal*.
Diagonal matrix	Square, with every off-diagonal entry zero.
Scalar / identity matrix	Diagonal with all diagonal entries equal (scalar), or all equal to 1 (identity, written I).
Zero (null) matrix	Every entry zero, of any order.
Upper / lower triangular	Square, with all entries below (upper) or above (lower) the diagonal equal to zero.
Symmetric	$a_{ij} = a_{ji}$ for all i, j — the matrix equals its transpose.
Skew-symmetric	$a_{ij} = -a_{ji}$; the diagonal entries are necessarily zero.
Orthogonal	A square matrix with $A\,A^{T} = I$, so its inverse equals its transpose.

2. Operations on matrices

- **Addition and subtraction** — only between matrices of the same order, entry by entry. Addition is commutative and associative.
- **Scalar multiplication** — every entry multiplied by the scalar.
- **Matrix multiplication** — the product AB exists only when the number of columns of A equals the number of rows of B. The entry $(AB)_{ij}$ is the dot product of row *i* of A with column *j* of B. Multiplication is associative and distributive, but **not commutative**: in general $AB \neq BA$, and $AB = 0$ does not imply $A = 0$ or $B = 0$.
- **Transpose** A^T — rows and columns exchanged. Key rules: $(A + B)^T = A^T + B^T$, $(AB)^T = B^T A^T$ (note the reversed order), $(kA)^T = kA^T$.

WORKED EXAMPLE — MULTIPLICATION

Let $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 5 \\ 0 & 2 \end{pmatrix}$. Then AB: entry (1,1) = 2·1 + 1·0 = 2; (1,2) = 2·5 + 1·2 = 12; (2,1) = 3·1 + 4·0 = 3; (2,2) = 3·5 + 4·2 = 23. So $AB = \begin{pmatrix} 2 & 12 \\ 3 & 23 \end{pmatrix}$. (BA would give a different matrix — check it as an exercise.)

3. Determinants

Every square matrix A has a number attached to it, its **determinant** |A| or det A. For 2 × 2, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has determinant $ad - bc$. For 3 × 3 and larger, the determinant is found by **cofactor expansion** along any row or column: each entry is multiplied by its cofactor — the determinant of the smaller matrix left after deleting its row and column, with the chessboard sign $(-1)^{i+j}$ — and the products are added.

WORKED EXAMPLE — A 3 × 3 DETERMINANT

Expand along the first row: | 1 2 3 ; 0 1 4 ; 5 6 0 | = 1·(1·0 − 4·6) − 2·(0·0 − 4·5) + 3·(0·6 − 1·5) = 1·(−24) − 2·(−20) + 3·(−5) = −24 + 40 − 15 = **1**.

Useful properties. Interchanging two rows changes the sign; a row of zeros, or two equal or proportional rows, makes the determinant zero; multiplying a row by k multiplies the determinant by k, so $\det(kA) = k^n \det A$ for order n; $\det(AB) = \det A \cdot \det B$; $\det(A^T) = \det A$; adding a multiple of one row to another leaves the determinant unchanged — the property that makes row reduction a practical way to evaluate large determinants.

4. Inverse of a matrix

A square matrix A is **non-singular** when $\det A \neq 0$, and then it possesses a unique **inverse** A^{-1} with $A\,A^{-1} = A^{-1}\,A = I$. If $\det A = 0$ the matrix is **singular** and no inverse exists. The inverse is built from the **adjugate** (the transpose of the cofactor matrix):

$$A^{-1} = (1 / \det A) \cdot \operatorname{adj} A$$

WORKED EXAMPLE — INVERTING A 2 × 2 MATRIX

$A = \begin{pmatrix} 4 & 7 \\ 2 & 6 \end{pmatrix}$; $\det A = 24 - 14 = 10$. For 2 × 2, swap the diagonal entries and change the signs of the off-diagonal ones: $\operatorname{adj} A = \begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$, so $A^{-1} = (1/10)\begin{pmatrix} 6 & -7 \\ -2 & 4 \end{pmatrix}$. Multiplying A by A^{-1} returns I, as it must.

5. Rank

The **rank** of a matrix is the order of the largest non-vanishing minor (sub-determinant), equivalently the number of independent (non-zero) rows after the matrix is reduced to row-echelon form by elementary operations. Rank measures how much independent information the matrix carries, and it decides the consistency of linear systems: the system AX = B has a solution exactly when rank(A) = rank(augmented matrix [A | B]), and the solution is unique when this common rank equals the number of unknowns.

6. Systems of linear equations

Three classical methods solve A X = B. **Matrix inversion** gives $X = A^{-1}B$ directly when A is non-singular. **Cramer's rule** writes each unknown as a ratio of determinants: $x_j = \det A_j / \det A$, where A_j is A with its j-th column replaced by B — elegant for small systems. **Gaussian elimination** reduces the augmented matrix to echelon form and back-substitutes; it is the method that scales to large systems and the one computers use.

WORKED EXAMPLE — CRAMER'S RULE

Solve $2x + 3y = 8$, $x - y = 1$. $D = (2)(-1) - (3)(1) = -5$. $D_x = (8)(-1) - (3)(1) = -11$, so $x = (-11)/(-5) = 2.2$. $D_y = (2)(1) - (8)(1) = -6$, so $y = (-6)/(-5) = 1.2$. Check: $2(2.2) + 3(1.2) = 8$ ✓ and $2.2 - 1.2 = 1$ ✓.

7. Eigenvalues and eigenvectors (introduction)

A non-zero vector X is an **eigenvector** of a square matrix A when $A X = \lambda X$ for some number λ: the matrix stretches the vector without turning it. The λ values are the **characteristic roots (eigenvalues)**, found from the characteristic equation $\det(A - \lambda I) = 0$. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0$ gives $\lambda = 1$ and $\lambda = 3$. Eigenvalues govern vibration, stability and stress problems throughout engineering, and return in later courses.

8. Key points and summary

- A matrix is an ordered array; operations respect order and shape — addition needs equal orders, multiplication needs matching inner dimensions, and $AB \neq BA$ in general.
- The determinant compresses a square matrix into one number: it tests singularity, scales with row operations by known rules, and multiplies over products.
- The inverse $A^{-1} = \text{adj } A / \det A$ exists exactly when $\det A \neq 0$, and solves systems by $X = A^{-1}B$. Rank decides consistency and uniqueness in general.
- Cramer’s rule, elimination and inversion are three views of the same solution; eigenvalues open the door to the matrix methods of later engineering courses.
